

Machine Learning Practical Assignment

Assignment: Decision Tree Regressor Using Kaggle Dataset

Total Marks: 10

Objective

Build a **Decision Tree Regressor** using a suitable dataset downloaded from **Kaggle**.

The objective of this assignment is to implement a Decision Tree Regression model, generate predictions, and evaluate the accuracy/performance of the model.

Assignment Task

You have to select **one suitable regression dataset from Kaggle** and build a **Decision Tree Regressor** using Python.

The dataset should contain a **numerical target variable** that can be predicted using the available features.

Examples:

- House Price Prediction
- Car Price Prediction
- Salary Prediction
- Sales Prediction
- Insurance Cost Prediction
- Student Score Prediction
- Electricity Consumption Prediction

You can select any other suitable regression dataset from Kaggle.

Steps to Perform

1. Select and Download Dataset

Download **one dataset from Kaggle**.

Mention the following in your submission:

- Dataset name
- Kaggle dataset link
- Number of rows and columns

- Target variable

2. Load and Understand the Dataset

Load the dataset using Python and perform basic data analysis.

Check:

- First few records
- Shape of the dataset
- Column names
- Data types
- Missing values

3. Prepare the Dataset

Identify:

- Input features (**X**)
- Target variable (**y**)

Perform only the necessary preprocessing required for your selected dataset.

4. Train-Test Split

Divide the dataset into training and testing data using a suitable train-test ratio.

5. Build the Model

Build **one Decision Tree Regressor** using the training data.

Use the trained model to predict the target values for the testing data.

6. Evaluate the Model

Evaluate your model using suitable regression metrics.

You must include:

- MAE
- MSE
- RMSE
- R^2 Score

7. Show the Output

Include the relevant outputs generated by your program.

Also include a suitable visualization such as:

- Actual vs Predicted Values

You may also include a visualization of the Decision Tree.

8. Conclusion

Write a short conclusion mentioning:

- Dataset used
- Target predicted
- Model performance
- R^2 Score
- One observation about your model

Submission Format

You have to submit your **code and output in a single Word document (.doc/.docx)**.

The document must contain:

1. **Student Name**
2. **Roll Number**
3. **Dataset Name**
4. **Kaggle Dataset Link**
5. **Python Code**
6. **Output Screenshots**
7. **Model Evaluation Results**
8. **Visualization Screenshot**
9. **Short Conclusion**

Important

The **Python code and its corresponding output must be clearly visible in the Word document**.

Take screenshots of the output wherever required and place them below the relevant code/output section.

You do **not** need to submit a separate report.

Marking Scheme – 10 Marks

Criteria	Marks
Correct implementation and model performance/accuracy	5
Code efficiency and minimum/appropriate lines of code	3
Output, screenshots and presentation	2

Criteria	Marks
Total	10 Marks

Note on Code Length

Students are expected to write **clean and concise code**.

Unnecessary code, repeated code, or unnecessary steps should be avoided.

Marks will be considered based on the **correctness of the implementation and how efficiently the required task is completed**, not simply on writing more lines of code.

Important Instructions

- Use **only one Kaggle dataset**.
- The problem must be a **regression problem**.
- Build **one Decision Tree Regressor**.
- The target variable must be numerical.
- Submit the work individually.
- The code must be your own and you should understand every part of it.
- You may be asked to explain your code and output during evaluation.
- Make sure all screenshots are clear and readable.
- The submitted Word document should contain both **code and output**.

Total Marks: 10

Submission Date: _____

Student Name: _____

Roll No.: _____